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Кожура опунции как ценный источник полисахаридов с добавленной стоимостью: исследование структурных, функциональных и пленкообразующих свойств

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Основные моменты

- Слизь из кожуры опунции содержит химические следы арабиногалактана.
- Раствор слизи в концентрации 4 % (по массе) показал псевдопластичные свойства.
- Слизистая оболочка кожуры показала интересные результаты в плане растворимости, пенообразования и эмульгирующих свойств.
- Пленки из слизистых оболочек пилинга обладают хорошей эластичностью и водоотталкивающими и жиротталкивающими свойствами.

Абстрактный

Повышение ценности побочных продуктов сельского хозяйства — перспективный подход к устойчивому развитию. Слизь была получена простым и безопасным способом из *Opuntia ficus-indica* кожуры, которая считается невостребованным отходом из-за огромного потребления опунции. Были изучены структурные и

функциональные свойства извлеченного полисахарида. Слизистая оболочка кожуры на 97 % состояла из углеводов, в основном галактозы, арабинозы, ксилозы и галактороновой кислоты. Специфические сигналы этих сахаров наблюдались в ^{13}C и ^1H спектрах ЯМР, а их химический состав был определен с помощью ИК-Фурье-спектроскопии с преобразованием Фурье в спектр. Измерения вязкости раствора слизи показали, что он разжижается при сдвиге. Выделенный биополимер обладал хорошей растворимостью в воде, пенообразующими и эмульгирующими свойствами. Благоприятная термическая стабильность проявлялась при температуре до 250 °C. Чтобы найти новые способы применения этого биополимера в качестве биоразлагаемой упаковки, были изучены его пленкообразующие свойства. Были получены пленки красного цвета с высоким углом смачивания водой ($\sim 91^\circ$), растворимостью ($\sim 42\%$) и жиростойкостью. Механические свойства были сопоставимы с характеристиками других полисахаридные пленки с приличным удлинением при разрыве, достигающим 66 %.

Ключевые слова

Слизь; Кожура опунции; Оценка стоимости отходов; Физико - химические свойства; Функциональные свойства; Съедобные пленки

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Введение

Кактус — широко распространённое растение в засушливых и полувасушливых регионах, таких как Америка и страны Средиземноморья, включая Тунис. Это быстрорастущая и недорогая культура, которая в основном используется в пищу (плоды и молодые кладодии) или в качестве корма (кладодии и кожура плодов) [1]. В Тунисе кактусами засажено около 600 000 гектаров [2]. Из-за огромного потребления опунции образуется большое количество неиспользуемых побочных продуктов и отходов, которые могут стать источником функциональных полимеров и соединений с добавленной стоимостью, в частности углеводов. Слизь — один из самых распространённых полисахаридов в кактусах. Она содержится в кожуре, мякоти и кладодиях плодов и достигает 4,1%, 3,8% и 19,4% соответственно для вида *Opuntia ficus-indica* [[3], [4], [5]]. Содержание слизистых веществ сильно зависит от вида кактуса, органа, возраста или степени зрелости органа, климатических условий (холод и дожди) и периода сбора урожая [5].

Cactus cladodes are succulent and articulate modified stems responsible for photosynthesis which takes place in the inner part of cladodes called collenchyma. The inside part made of the white medullar parenchyma, is responsible for water storage. The mucilaginous cells are distributed in both parts but are more abundant in parenchyma [5]. Cladodes are eaten as a vegetable in some countries but in other ones, they are considered by-products. Cactus cladodes mucilage may have several applications in food, pharmaceutical and cosmetic industries as thickening, gelling and emulsifying agents [6,7], encapsulating agent [1], heavy metal-removing agent [8], film forming matrices [9] and it has been recently proposed as a mucoprotective and dermal wound healing agent [10].

Fruit peels, also known as fruit skins, are considered as wastes resulting from the huge consumption of the fresh fruit. These peels are also characterized by collenchyma and parenchyma tissues containing mucilaginous cells [3]. In many countries, the peels are either discarded or used for animal feeding. It is imperative to develop effective strategies to valorize these peel wastes in order to produce high added value products. Although most studies on cactus mucilage have focused on mucilage extracted from cladodes, only few studies have attempted to extract polysaccharides from peels. Pectic polysaccharides from *O. ficus-indica* were characterized by a branched structure with a compact sphere conformation, composed of galacturonic acid and rhamnose [11]. Recently, Han et al. [12] studied the effect of microwave-assisted extraction parameters on viscosity, microstructure and thermal stability of peels mucilage from *O. dillenii haw.* species.

Nowadays, due to the increasing global demand for plastic materials associated with several negative effects on the environment and human health, the demand for new innovative and eco-friendly materials, commonly called bioplastics, is emerging and alternative food stocks constitute interesting option for the development of bioplastics [13]. In addition, the increasing global need for food may improve the development of bioplastics from non-food products [13]. In this current context, cactus mucilage may be an interesting biopolymer for the development of eco-friendly materials and added-value products. Indeed, mucilage aqueous solutions showed high elastic properties similar to those of high elastic synthetic polymers [7]. Moreover, biopolymer-based materials are preferred over synthetic ones due to their natural availability, biocompatibility, biodegradability and non-toxicity [6].

Therefore, the objective of this study was to extract and characterize mucilage from *O. ficus-indica* fruit peels and process it into films to ultimately evaluate their application as bioplastics for packaging materials.

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Plant material

Orange-red colored prickly pears from *O. ficus-indica* species were freshly harvested in the period of June 2016 in the region of Thala located in the west of Tunisia (latitude of 35°34'N; longitude of 8°40' E). Fruits were selected in the same stage of ripeness and without physical or microbial damage on the skin. ...

Mucilage extraction

Mucilage was extracted from fruit peels following the method described by Gheribi et al. [9]. Fresh pears were washed and hand peeled. The peels were mechanically pressed and the ...

Extraction yield

Prickly pear peels constitute a widespread and low cost waste that can be usefully valorized as a source of valuable biopolymers. Since polysaccharides from natural resources find huge interest in cosmetic and pharmaceutical industries, mucilage from fruit peels may be an interesting biopolymer with many potential applications. Mucilage extraction yield from *O. ficus-indica* fruit peels was 3.08% (% dry weight) and seems in agreement with results found by Habibi et al. [3] for the same variety. ...

Conclusion

Prickly pear peels mucilage exhibited promising functional properties which can be exploited in several industrial applications. Indeed, peels mucilage can act as natural emulsifier, foaming agent or water holder in food industries and cosmetics. The incontestable film forming character of this biopolymer, when plasticized with glycerol, makes it suitable to

serve as coating materials for foodstuffs or paper and cardboard packaging materials thanks to its safety and non-toxicity. The film ...

Acknowledgments

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de Andrade Vieira É., ..., Cordeiro Â.M.T.D.M.

Food Chemistry • Volume 346 • 2021 • Article 128892

Citation Excerpt:

...The *Opuntia dillenii* mucilage showed an emulsifying capacity (EC) higher than 60% and emulsifying stability (ES) up to 50% (Kalegowda, Chauhan, & Nanjaraj Urs, 2017). *Opuntia ficus-indica* mucilage had

emulsions capacity of 48.89% and emulsification stability of 45.71% (Gheribi, Habibi, & Khwaldia, 2019). Dick et al. (2019) found that the EC of *Opuntia monacantha* mucilage was greater than 90%, with ES above 95%....

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Review article

Recent advances in the sustainable design and applications of biodegradable polymers

Rai P., ..., Sharma S.K.

Bioresource Technology • Volume 325 • 2021 • Article 124739

Citation Excerpt:

...The valorized biomass is 75% polysaccharides and can be used to develop stand-alone and blended polymeric matrix for use as edible packaging. Stand-alone edible coating of mucilage require plasticizers such as glycerol (Gheribi et al., 2019b), sorbitol, PEG (Gheribi et al., 2018) and PVA (Dominguez-Martinez et al., 2017) to improve their tensile strength and thermal endurance. As a blended polymer, mucilage along with starch (López-García et al., 2017), chitosan (Gheribi et al., 2019a) and beeswax (Lira-Vargas et al., 2014) has been used to prepare edible food coating with thermal and mechanical stability and barrier properties towards water and gases....

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Kamel R., ..., Farag M.A.

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